

## CHAPTER II.7, WEEK 9

Noah Parker

- 7.1. Let  $(X, \mathcal{O}_X)$  be a locally ringed space, and let  $f : \mathcal{L} \rightarrow \mathcal{M}$  be a surjective map of invertible sheaves on  $X$ . Show that  $f$  is an isomorphism. [*Hint*: Reduce to a question of modules over a local ring by looking at the stalks.]
- 7.2. Let  $X$  be a scheme over a field  $k$ . Let  $\mathcal{L}$  be an invertible sheaf on  $X$ , and let  $\{s_0, \dots, s_n\}$  and  $\{t_0, \dots, t_m\}$  be two sets of sections of  $\mathcal{L}$ , which generate the same subspace  $V \subset \Gamma(X, \mathcal{L})$ , and which generate the sheaf  $\mathcal{L}$  at every point. Suppose  $n \leq m$ . Show that the corresponding morphisms  $\phi : X \rightarrow \mathbb{P}_k^n$  and  $\psi : X \rightarrow \mathbb{P}_k^m$  differ by a suitable linear projection  $\mathbb{P}^m - L \rightarrow \mathbb{P}^n$ , and an automorphism of  $\mathbb{P}^n$ , where  $L$  is a linear subspace of dimension  $m - n - 1$ .
- 7.3. Let  $\phi : \mathbb{P}_k^n \rightarrow \mathbb{P}_k^m$  be a morphism. Then:
- (a) either  $\phi(\mathbb{P}^n) = \{*\}$  or  $m \geq n$  and  $\dim \phi(\mathbb{P}^n) = n$ ;
  - (b) in the second case,  $\phi$  can be obtained as the composition of (1) a  $d$ -uple embedding  $\mathbb{P}^n \rightarrow \mathbb{P}^N$  for a uniquely determined  $d \geq 1$ , (2) a linear projection  $\mathbb{P}^N - L \rightarrow \mathbb{P}^m$ , and (3) an automorphism of  $\mathbb{P}^m$ .
- 7.5. Establish the following properties of ample and very ample invertible sheaves on a noetherian scheme  $X$ .  $\mathcal{L}, \mathcal{M}$  will denote invertible sheaves, and for (d), (e) we assume furthermore that  $X$  is of finite type over a noetherian ring  $A$ .
- (a) If  $\mathcal{L}$  is ample and  $\mathcal{M}$  is generated by global sections, then  $\mathcal{L} \otimes \mathcal{M}$  is ample.
  - (b) If  $\mathcal{L}$  is ample and  $\mathcal{M}$  is arbitrary, then  $\mathcal{M} \otimes \mathcal{L}^n$  is ample for  $n \gg 0$ .
  - (c) If  $\mathcal{L}, \mathcal{M}$  are both ample, so is  $\mathcal{L} \otimes \mathcal{M}$ .
  - (d) If  $\mathcal{L}$  is very ample and  $\mathcal{M}$  is generated by global sections, then  $\mathcal{L} \otimes \mathcal{M}$  is very ample.
  - (e) If  $\mathcal{L}$  is ample, then there is an  $n_0 > 0$  such that  $\mathcal{L}^n$  is very ample for all  $n \geq n_0$ .
- 7.7. *Some Rational Surfaces.* Let  $X = \mathbb{P}_k^2$ , and let  $|D|$  be the complete linear system of all divisors of degree 2 on  $X$  (conics).  $D$  corresponds to the invertible sheaf  $\mathcal{O}(2)$ , whose space of global sections has a basis  $x^2, y^2, z^2, xy, xz, yz$ , where  $x, y, z$  are the homogeneous coordinates of  $X$ .
- (a) The complete linear system  $|D|$  gives an embedding of  $\mathbb{P}^2$  in  $\mathbb{P}^5$ , whose image is the veronese surfaces (I, Ex. 2.13).
  - (b) Show that the subsystem defined by  $x^2, y^2, z^2, y(x - z), (x - y)z$  gives a closed immersion of  $X$  into  $\mathbb{P}^4$ . The image is called the *Veronese surface* in  $\mathbb{P}^4$ . Cf. (IV, Ex. 3.11).
  - (c) Let  $\mathfrak{d} \subset |D|$  be the linear system of all conic passing through a fixed point  $p$ . Then  $\mathfrak{d}$  gives an immersion of  $U = X - p$  into  $\mathbb{P}^4$ . Furthermore, if we blow up  $p$  to get a surface  $\tilde{X}$ , then this map extends to give a closed immersion of  $\tilde{X}$  in  $\mathbb{P}^4$ . Show that  $\tilde{X}$  is a surface of degree 3 in  $\mathbb{P}^4$ , and that the lines in  $X$  through  $p$  are transformed into straight lines in  $\tilde{X}$  which do not meet.  $\tilde{X}$  is the union of all these lines, so we say  $\tilde{X}$  is a *ruled surface* (V, 2.19.1).